

Chapter 1

Operations and Maintenance Costs Documentation

1.1 Introduction

The `oandm.py` script calculates operations and maintenance (O&M) costs for a transmission line project. It computes annual and lifetime costs for conductors, converters, structures, and vegetation management, then calculates present values.

1.2 Method Overview

Canonical notation. The canonical variable names and equations for O&M are defined in `COST_METHODODOLOGY.md` under “Operational costs” / “O&M (2.a)”. Structure cost is structure-only and vegetation is a separate component: $C_{\text{structure,annual}}$ is piecewise (overhead vs. non-overhead), $C_{\text{veg,annual}} = \sum_{\text{terrain}} M_{\text{terrain}} \times c_{\text{veg}}(\text{terrain})$, and $C_{\text{O\&M,annual}} = C_{\text{conductor,annual}} + C_{\text{converter,annual}} + C_{\text{structure,annual}} + C_{\text{veg,annual}}$.

1.2.1 Purpose

This script calculates O&M costs by:

1. Loading O&M cost parameters by component (conductor, converter, structure, vegetation)
2. Calculating annual costs per component
3. Computing lifetime nominal costs
4. Calculating present values of annual O&M payments

1.2.2 Calculation Approach

The O&M cost calculation follows this structure. Structure Annual is structure-only (vegetation is a separate line):

$$\text{Conductor Annual} = \text{Cost/Mile/Year} \times \text{Total Miles} \quad (1.1)$$

$$\text{Converter Annual} = \text{Cost/Mile/Year} \times \text{Total Miles} \quad (\text{DC only}) \quad (1.2)$$

$$\text{Structure Annual (structure only)} = \begin{cases} \text{Cost/Structure/Year} \times \text{Total Structures} & (\text{Overhead}) \\ \text{Cost/Mile/Year} \times \text{Total Miles} & (\text{Other}) \end{cases} \quad (1.3)$$

$$\text{Vegetation Annual} = \sum_{\text{terrains}} \text{Cost/Mile/Year}_{\text{terrain}} \times \text{Miles}_{\text{terrain}} \quad (1.4)$$

$$\text{Total Annual} = \text{Conductor} + \text{Converter} + \text{Structure} + \text{Vegetation} \quad (1.5)$$

$$\text{Lifetime Nominal} = \text{Annual} \times \text{Project Lifetime} \quad (1.6)$$

$$\text{Present Value} = \sum_{t=1}^N \frac{\text{Annual}}{(1+r)^t} \quad (1.7)$$

1.2.3 Inputs and Outputs

Inputs:

- Project technical details (construction type, AC/DC, capacity, conductor type, converter type)
- Physical details (total miles, terrain distribution)
- O&M cost parameters (conductor, converter, structure, vegetation management)
- Financing parameters (WACC for present value)
- Project lifetime

Outputs:

- Annual costs by component
- Lifetime nominal costs by component
- Present value costs by component
- Total annual, nominal, and PV costs

1.2.4 Integration with Other Scripts

This script integrates with:

- `yaml_loaders.py` — `load_financing_details`, `load_project_technical_details`, `load_physical_load_circuit_and_resistance_details`

- `financial_utils.py` — `calculate_present_value`, `calculate_cod_year`
- `calculation_utils.py` — `build_category_string`
- `path_config.py` — `YAMLS_DIR`
- `constants.py` — `CONSTRUCTION_TYPE_OVERHEAD`
- `smart_output.py` — `CTCCOutputManager` (`add_oandm_costs`, `write_batch_summary`)

1.3 Key Functions

1.3.1 `load_vegetation_management_om_costs(construction_type)`

Loads vegetation management O&M costs from YAML file.

Parameters:

- `construction_type` (str): Type of construction (Overhead/Subsea)

Returns:

- Dict mapping terrain key (e.g. `forested`, `farmland`) to cost per mile per year

1.3.2 `load_conductor_om_costs(construction_type, ac_dc, capacity_mw, conductor_type, converter_type)`

Loads conductor O&M costs from YAML file.

Parameters:

- `construction_type` (str): Type of construction
- `ac_dc` (str): AC or DC designation
- `capacity_mw` (float): Capacity in megawatts
- `conductor_type` (str): Type of conductor
- `converter_type` (str): Type of converter

Returns:

- `variable_conductor_cost_per_mile_year` (float): Variable conductor cost per mile per year

1.3.3 load_converter_om_costs(construction_type, ac_dc, capacity_mw, conductor_type, converter_type)

Loads converter O&M costs from YAML file. Returns 0 for AC projects.

Parameters:

- `construction_type` (str): Type of construction
- `ac_dc` (str): AC or DC designation
- `capacity_mw` (float): Capacity in megawatts
- `conductor_type` (str): Type of conductor
- `converter_type` (str): Type of converter

Returns:

- `variable_converter_cost_per_mile_year` (float): Variable converter cost per mile per year (0 for AC)

1.3.4 load_structure_om_costs(construction_type, forested_miles, ...)

Loads structure O&M from `14.category_om_structures.yaml`. For overhead: total structures by terrain (`structures_per_mile_forested`, etc.), cost per structure per year, then adds vegetation management (terrain-specific \$/mile/year from `load_vegetation_management_om_costs`). For non-overhead: returns `variable_cost_per_mile_year` only.

Parameters:

- `construction_type`; then `forested_miles`, `scrubbed_flat_miles`, `wetland_miles`, `farmland_miles`, `desert_barren_miles`, `urban_miles`, `rolling_hills_miles`, `mountain_miles`, `subsea_miles`

Returns:

- `variable_structure_cost_per_mile_year` (float), `variable_structure_cost_per_year` (float), `structure_dict` (counts by terrain), `total_vegetation_management_cost_per_year` (float)

Key Logic for Overhead:

```
1 category = construction_type # structure YAML keyed by construction
   type
2 # structures_per_mile_forested, structures_per_mile_farmland, ...
   from YAML
3 forested_structures = forested_miles * structures_per_mile_forested
4 # ... same for each terrain; subsea_structures = 0
5 total_structures = sum(terrain_structures)
6 variable_structure_cost_per_year = cost_per_structure_per_year *
   total_structures
```

```

7 vegetation_management_cost_per_mile_year =
  load_vegetation_management_om_costs(construction_type)
8 # per terrain: miles * vegetation_management_cost_per_mile_year[
  terrain]
9 total_vegetation_management_cost_per_year = sum(...)
10 variable_structure_cost_per_year +=
  total_vegetation_management_cost_per_year

```

1.3.5 main()

Main execution function that orchestrates the O&M cost calculation and output.

Workflow:

1. Loads project technical details and physical details
2. Loads O&M cost parameters for each component
3. Calculates annual costs per component
4. Calculates lifetime nominal costs
5. Calculates present values
6. Displays results and writes to CSV

1.4 Example

The following example demonstrates a typical usage of the O&M cost calculation:

```

1 # Example: 500 MW Overhead AC line
2 # Category: "Overhead/AC/500MW/Advanced Aluminum/NA"
3 #
4 # Inputs (from YAML):
5 #   - Total miles: 100 miles
6 #   - Terrain: 60% farmland, 40% forested
7 #   - Conductor O&M: $5,000/mile/year
8 #   - Converter O&M: $0 (AC project)
9 #   - Structure O&M: $500/structure/year
10 #   - Structures: 2.5/mile (farmland), 3.0/mile (forested)
11 #   - Vegetation: $2,000/mile/year (farmland), $3,000/mile/year (
    forested)
12 #   - Project lifetime: 50 years
13 #   - Real WACC: 3%
14 #
15 # Calculation:
16 #   Conductor annual = $5,000 * 100 = $500,000/year
17 #   Converter annual = $0
18 #   Total structures = (60 * 2.5) + (40 * 3.0) = 270 structures

```

```

19 # Structure annual = $500 * 270 = $135,000/year
20 # Vegetation annual = (60 * $2,000) + (40 * $3,000) = $240,000/
    year
21 # Total annual = $875,000/year
22 # Lifetime nominal = $875,000 * 50 = $43.75M
23 # PV = $875,000 * PV annuity (50 years, 3%) = $20.1M
24 #
25 # Output:
26 #   - Total annual: $875,000
27 #   - Lifetime nominal: $43.75M
28 #   - Present value: $20.1M

```

Integration Example:

```

1 # When called from ctcc.py:
2 csv_manager = CTCCOutputManager()
3 results = {
4     "total_annual": total_annual,
5     "total_nominal": total_nominal,
6     "total_pv": pv_total,
7     "conductor_annual": total_conductor_cost_per_year,
8     # ... component-level costs
9 }
10 csv_manager.add_oandm_costs(results)
11 csv_manager.write_batch_summary()

```

1.5 Key Implementation Details

1.5.1 Structure O&M for Overhead Projects

For overhead projects, structure O&M is calculated based on the number of structures, which varies by terrain type. The script:

1. Calculates structures per terrain based on structures per mile
2. Sums to get total structures
3. Multiplies by cost per structure per year

1.5.2 Vegetation Management

Vegetation management costs are terrain-specific and only apply to overhead projects. The script multiplies terrain-specific costs per mile by miles in each terrain type.

1.5.3 Converter O&M

Converter O&M costs only apply to DC projects. For AC projects, the function returns 0.

1.5.4 Present Value Timing

O&M costs start when the project becomes operational (after delay and construction). The script uses `calculate_cod_year(delay_years, construction_years)` as `start_year` in `calculate_present_value` to discount annual O&M payments back to the base year.

1.5.5 Component-Level Tracking

The script tracks costs at the component level (conductor, converter, structure, vegetation) for detailed analysis and reporting.